

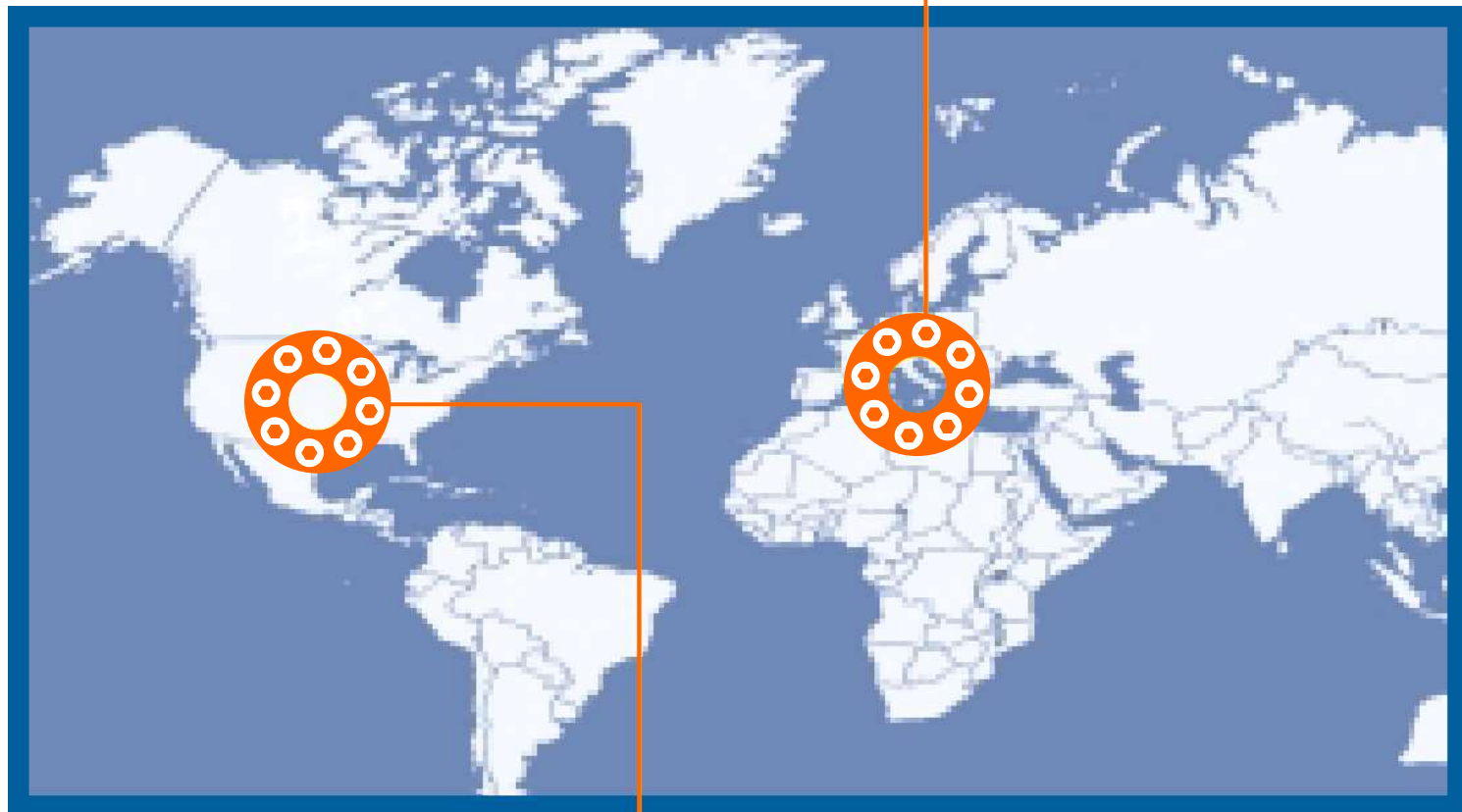
TOLLOK®

KEYLESS

LOCKING ASSEMBLIES



Connecting Technology . . .



**Tollok Europe
(Ferrara, Italy)**

**Tollok North America
(Atlanta, GA)**

. . . with Local Support

Connecting Technology . . .

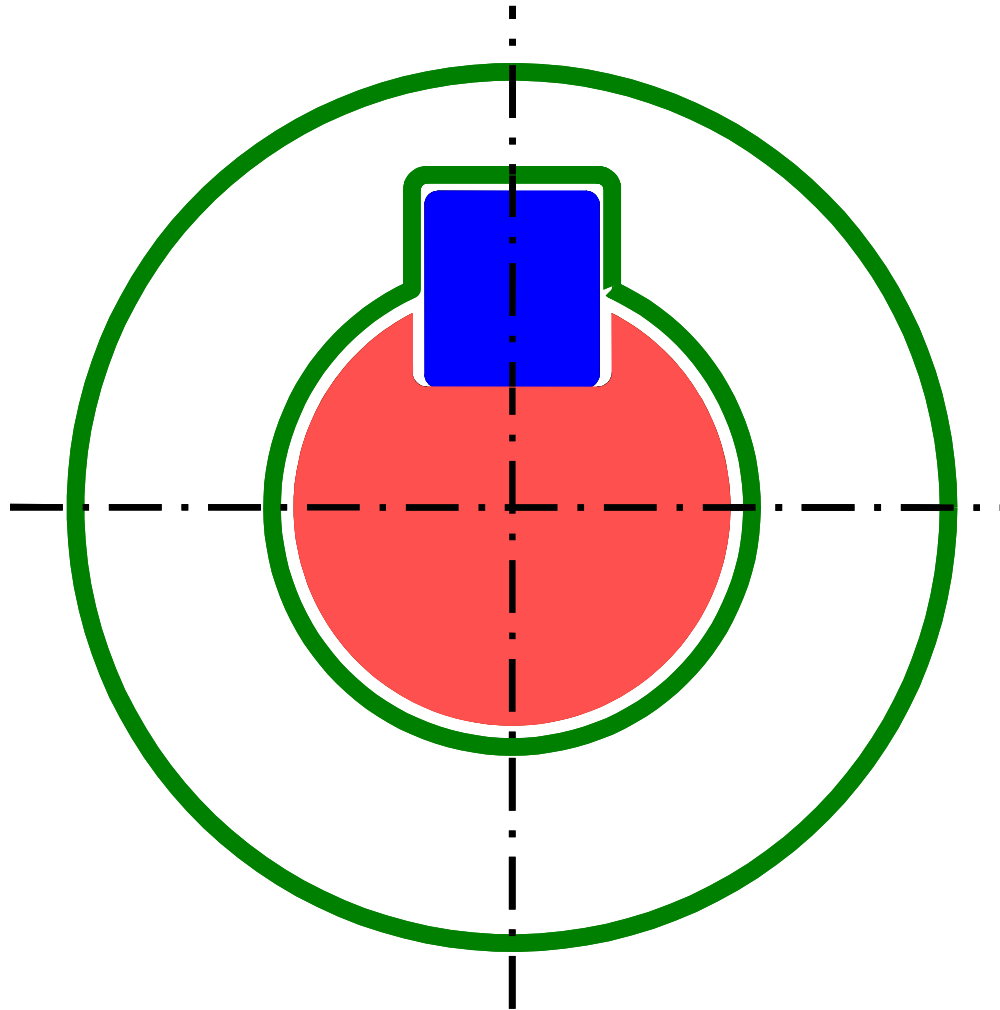


. . . with Quality products

DISADVANTAGES OF USING CONVENTIONAL SHAFT/HUB CONNECTIONS SUCH AS . . .

- ◆ **KEYED CONNECTIONS**
- ◆ **SPLINED CONNECTIONS**
- ◆ **TAPER-LOCKS**
- ◆ **QD BUSHINGS**
- ◆ **INTERFERENCE FIT**

THROW AWAY THE KEY



KEYED SHAFT / HUB CONNECTIONS

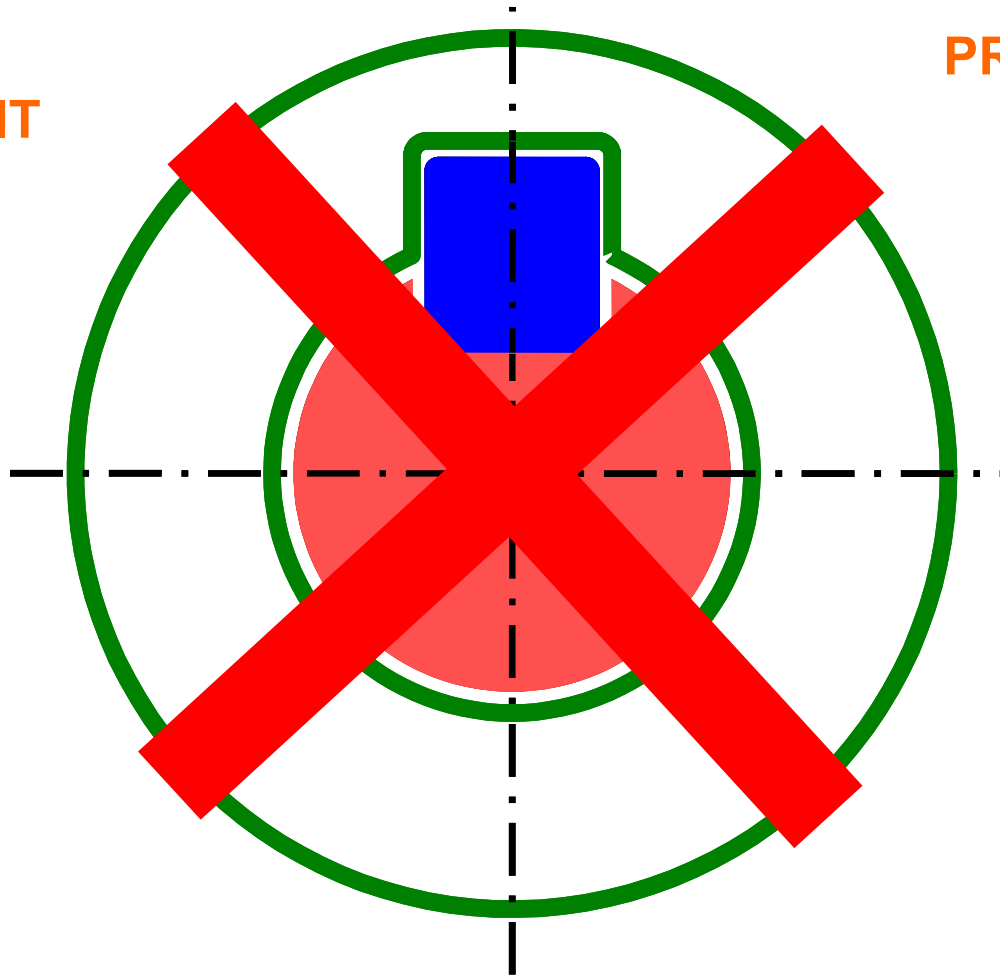
DISADVANTAGES OF...

SMALL SURFACE
AREA TO TRANSMIT
TORQUE

PRONE TO FRETTING
CORROSION

HIGH LEVEL OF
STRESS
CONCENTRATION
IN SMALL AREA

DIFFICULT TO
DISASSEMBLE



KEYED SHAFT / HUB CONNECTIONS

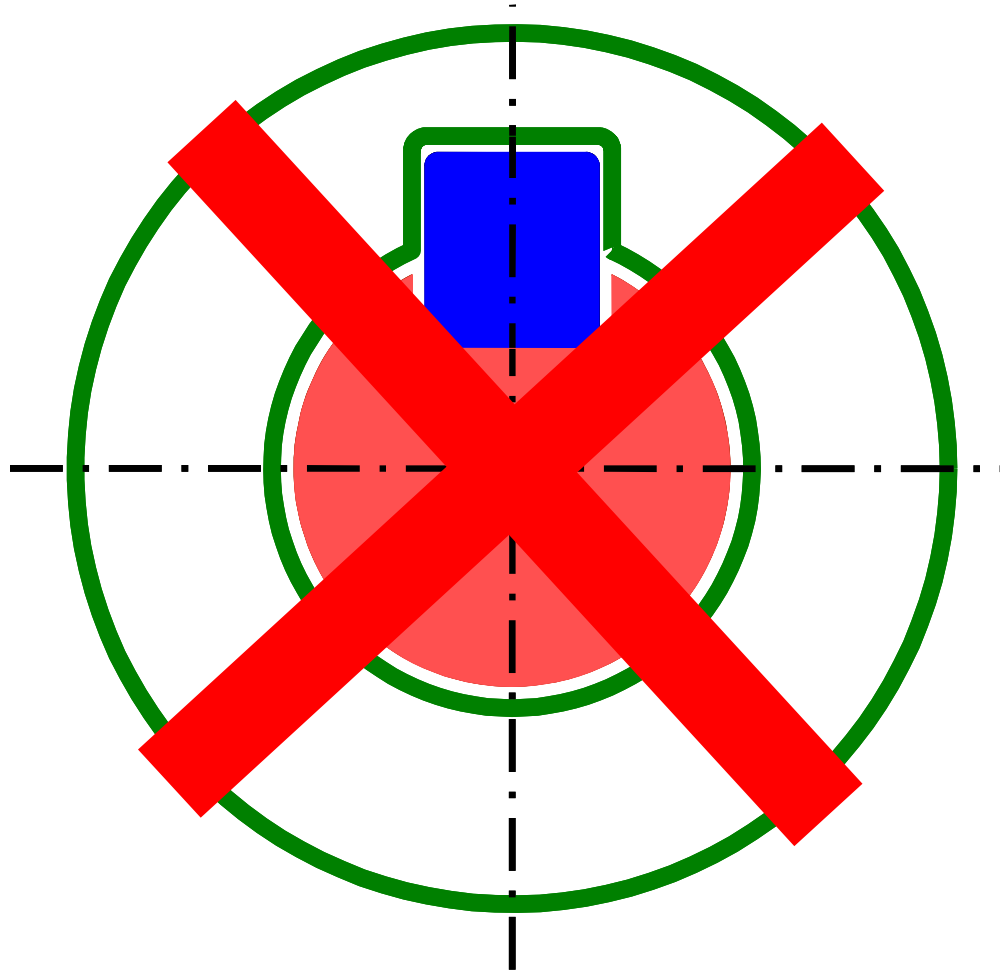
DISADVANTAGES OF...

SINGLE POSITION
CAPABILITY

REQUIRES USE OF
LARGER SHAFT &
HUB DIAMETERS

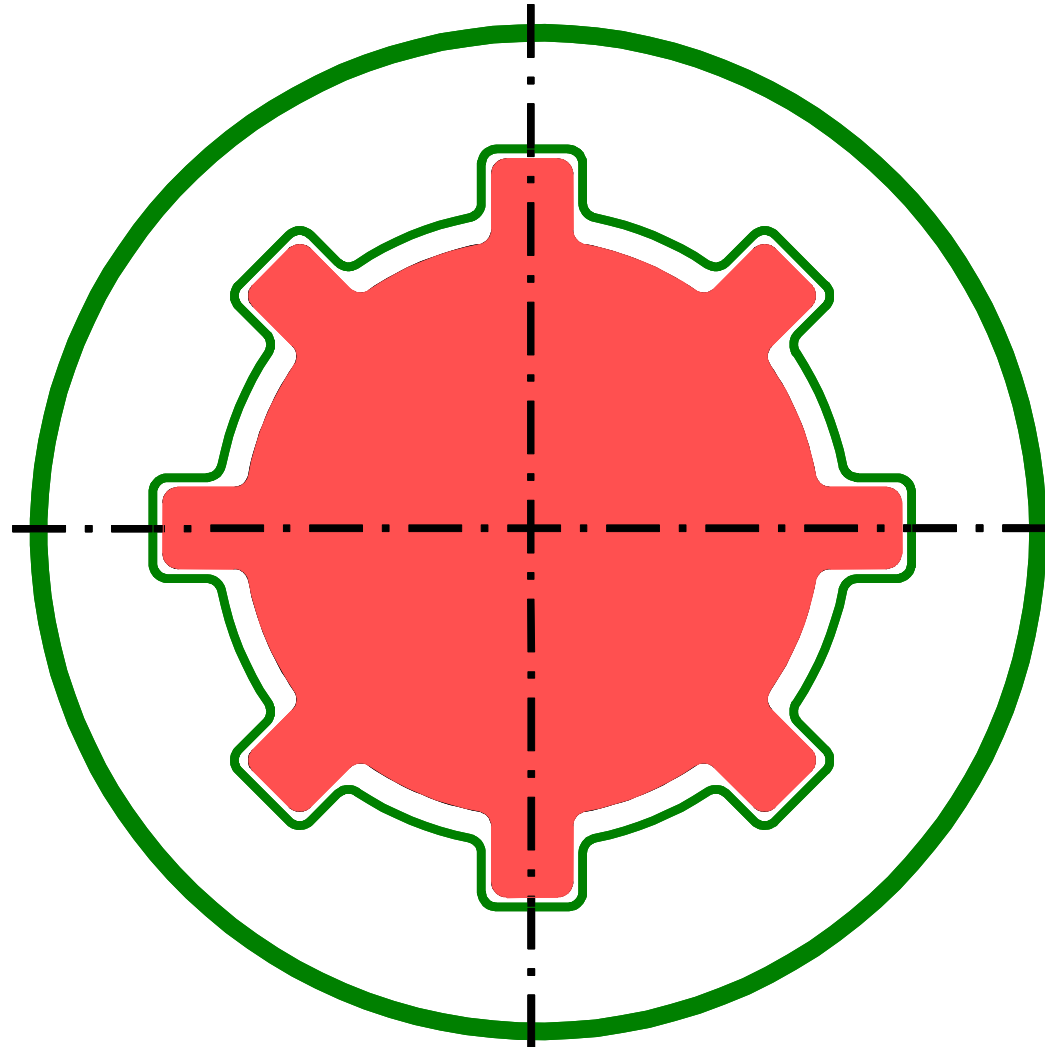
ADDS BACKLASH
IN BIDIRECTIONAL
SHAFT/HUB
ROTATION

POOR SOLUTION
FOR HIGH LEVELS
OF SHOCK LOADS



KEYED SHAFT / HUB CONNECTIONS

WHY NOT TO USE



SPLINED SHAFT / HUB CONNECTIONS

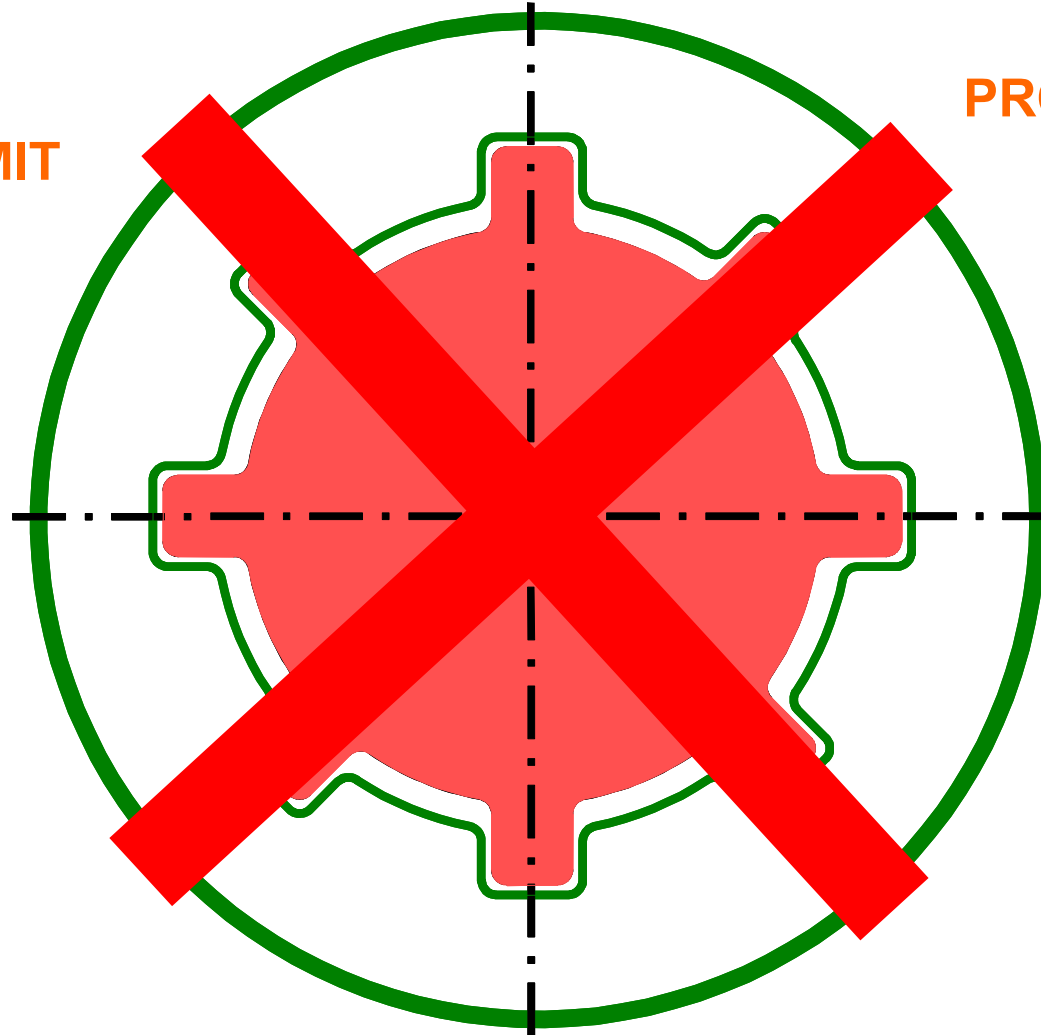
DISADVANTAGES OF...

SMALL SURFACE
AREA TO TRANSMIT
TORQUE

PRONE TO FRETTING
CORROSION

HIGH LEVEL OF
STRESS
CONCENTRATION
IN SMALL AREA

DIFFICULT TO
DISASSEMBLE



SPLINED SHAFT / HUB CONNECTIONS

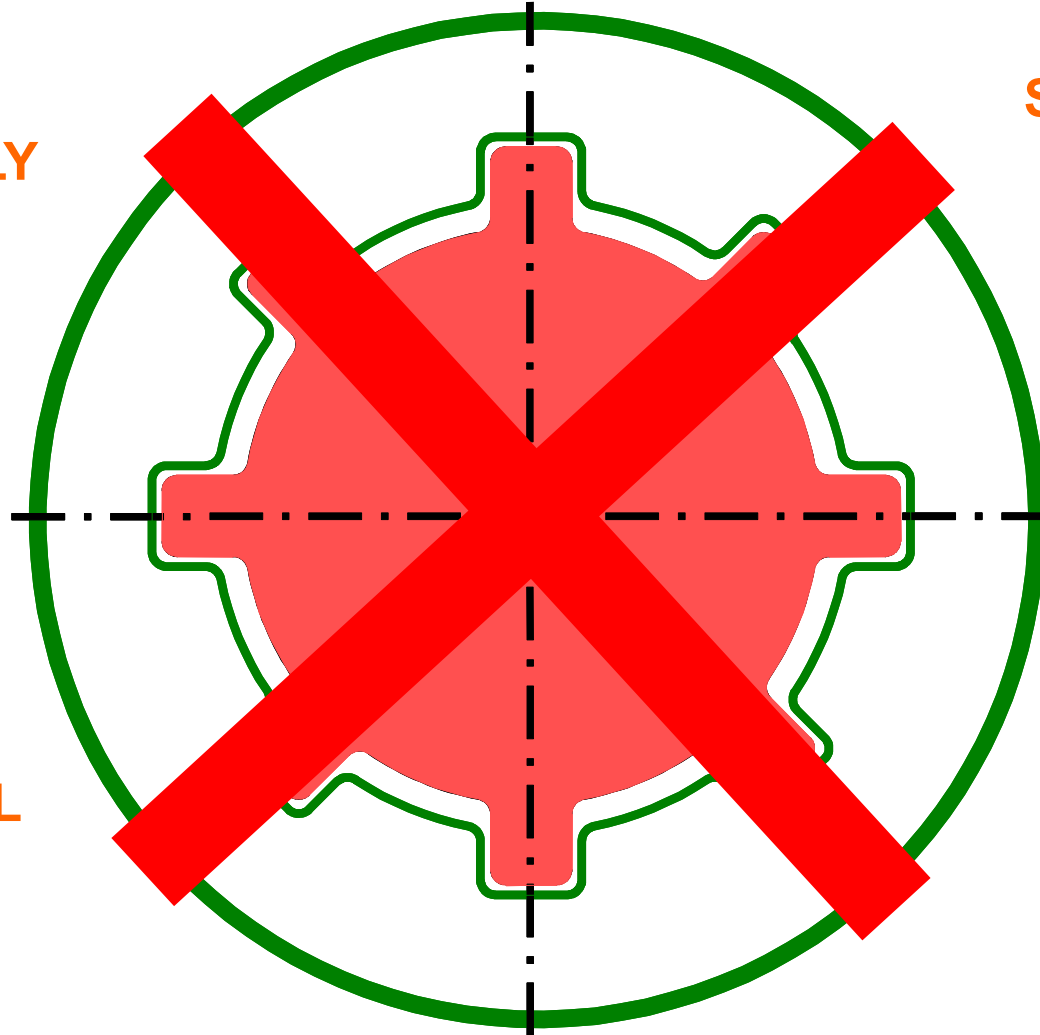
DISADVANTAGES OF...

HIGH MACHINING
COST, ESPECIALLY
HOLLOW SHAFTS

SPECIAL SURFACE
TREATMENT IS
REQUIRED TO
INCREASE JOINT
STRENGTH

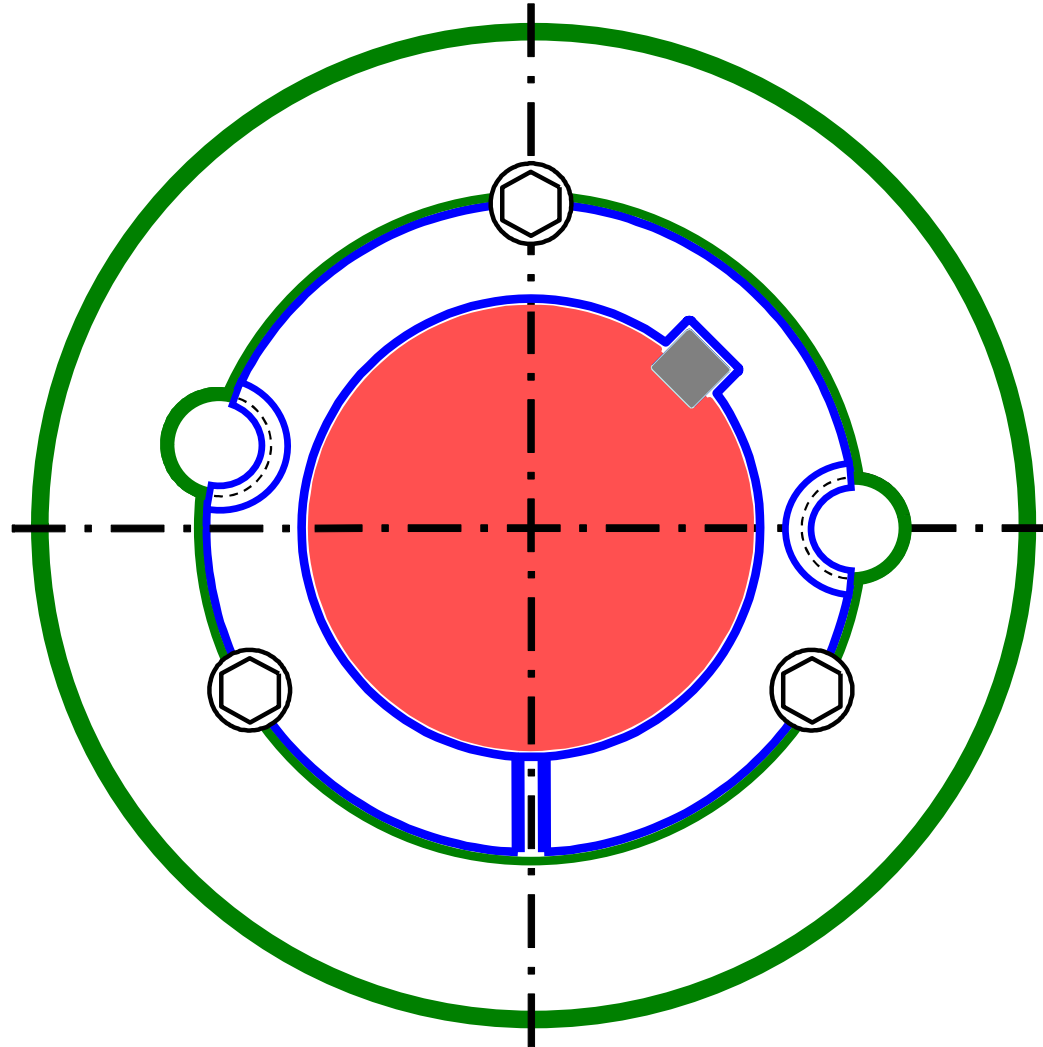
ADDS BACKLASH
IN BIDIRECTIONAL
SHAFT/HUB
ROTATION

DIFFICULT TO
REPAIR IN THE
FIELD



SPLINED SHAFT / HUB CONNECTIONS

WHY NOT TO USE



TAPER-LOCK SHAFT / HUB CONNECTIONS

DISADVANTAGES OF...

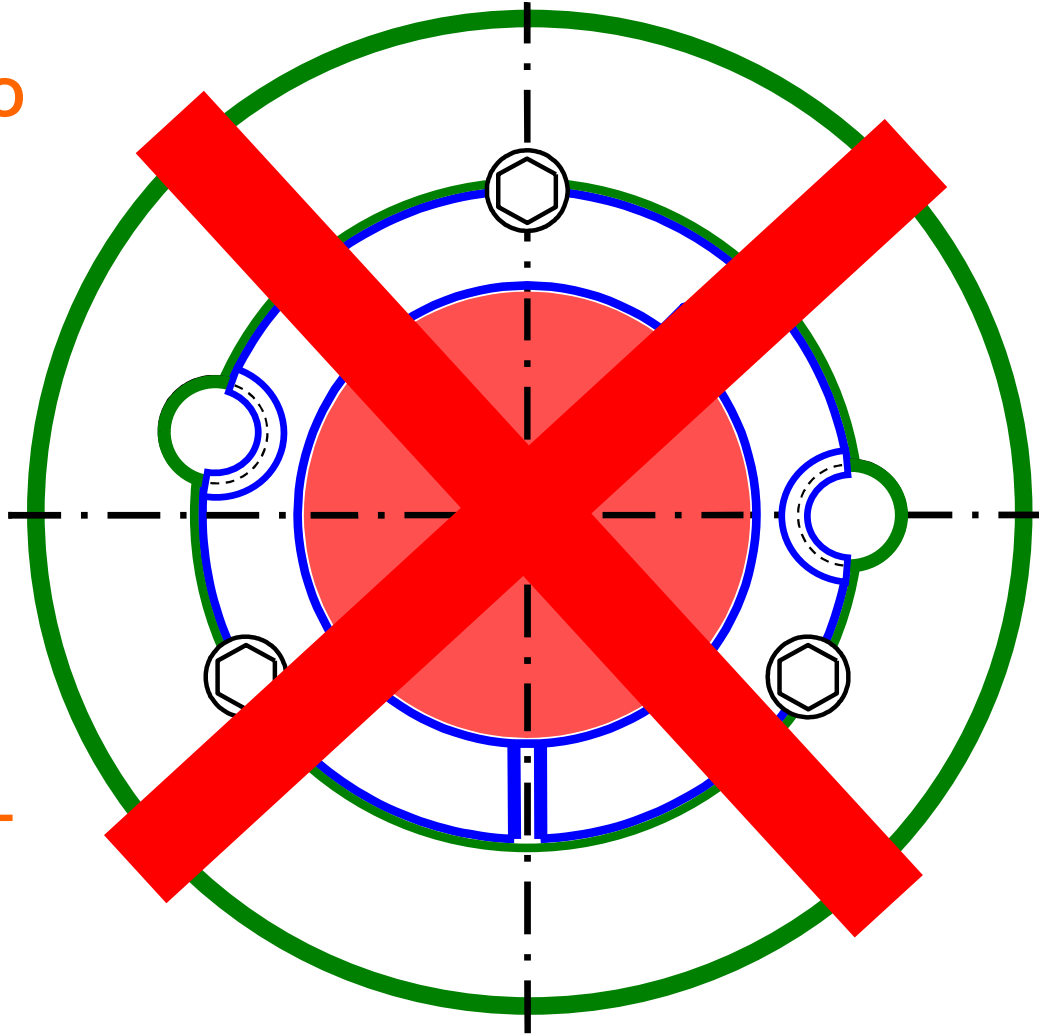
REQUIRES HUB TO
HAVE MATCHING
TAPERS AND
THREADS

HIGH MACHINING
COST DUE TO
TAPER AND
THREADING

ADDS BACKLASH
IN BIDIRECTIONAL
SHAFT/HUB
ROTATION

SINGLE POSITION
CAPABILITY

TAPER-LOCK SHAFT / HUB CONNECTIONS



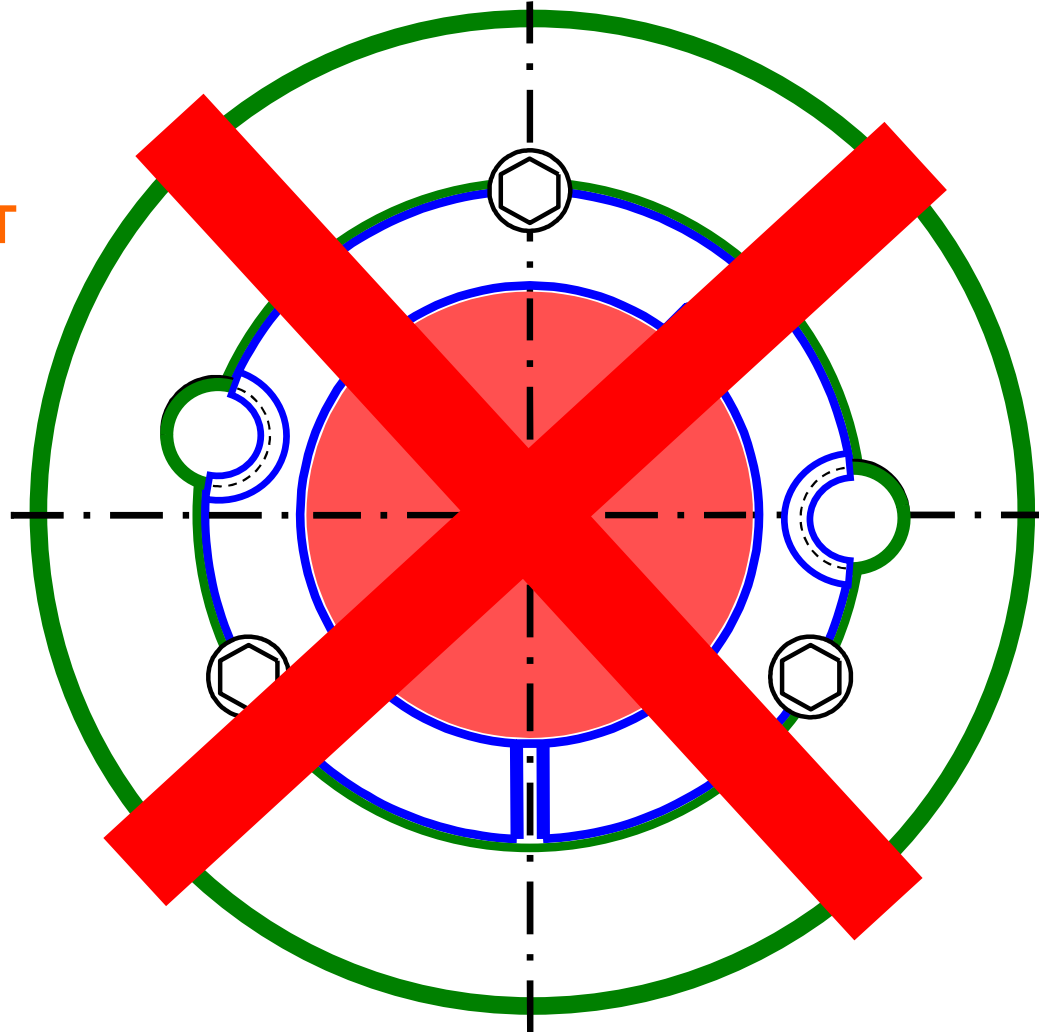
DISADVANTAGES OF...

REQUIRES
SECONDARY
LOCKING (e.g. SET
SCREWS, KEY)

REQUIRES NON-
STANDARD
COMPONENTS

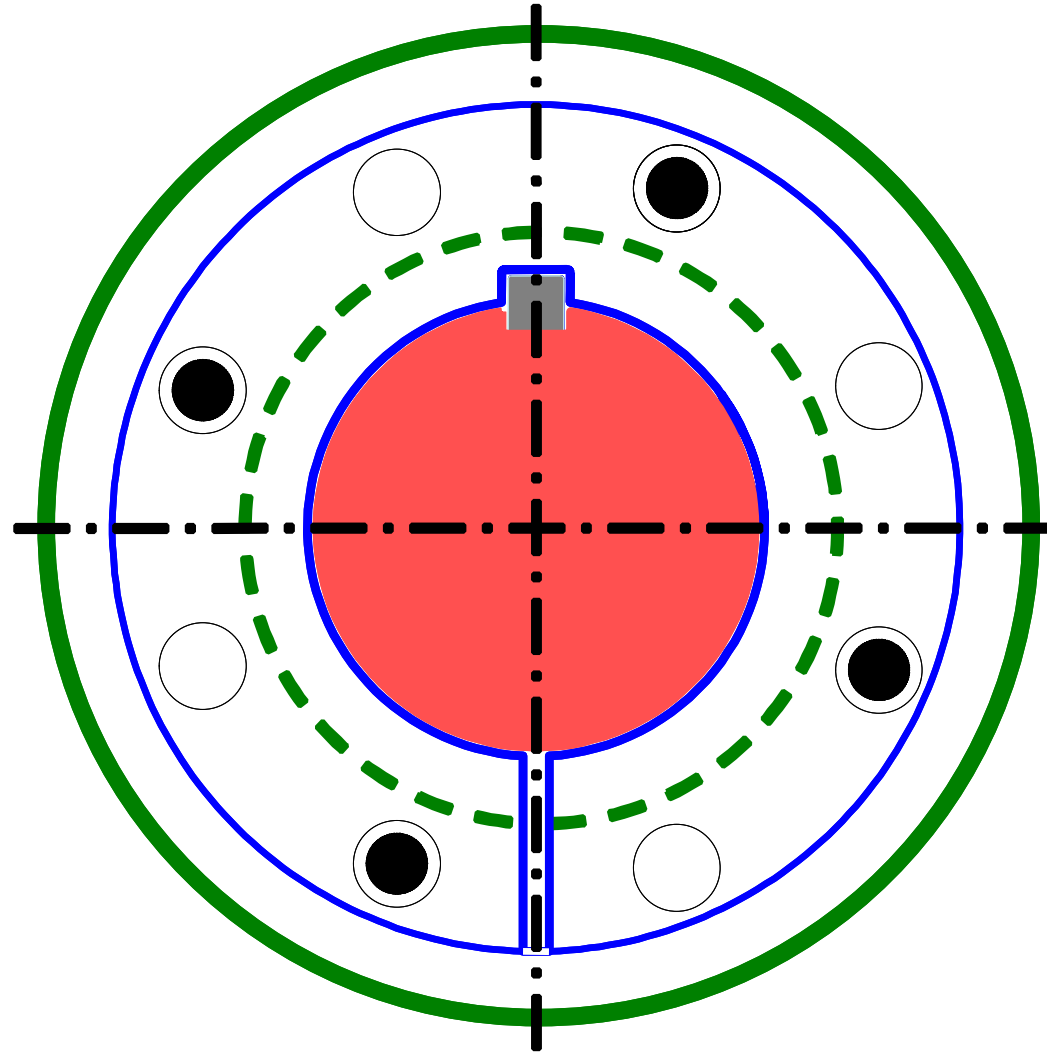
DISASSEMBLY
REQUIRES
ADDITIONAL
HARDWARE

LOCALIZED
STRESS
CONCENTRATION



TAPER-LOCK SHAFT / HUB CONNECTIONS

WHY NOT TO USE



QD BUSHING SHAFT / HUB CONNECTIONS

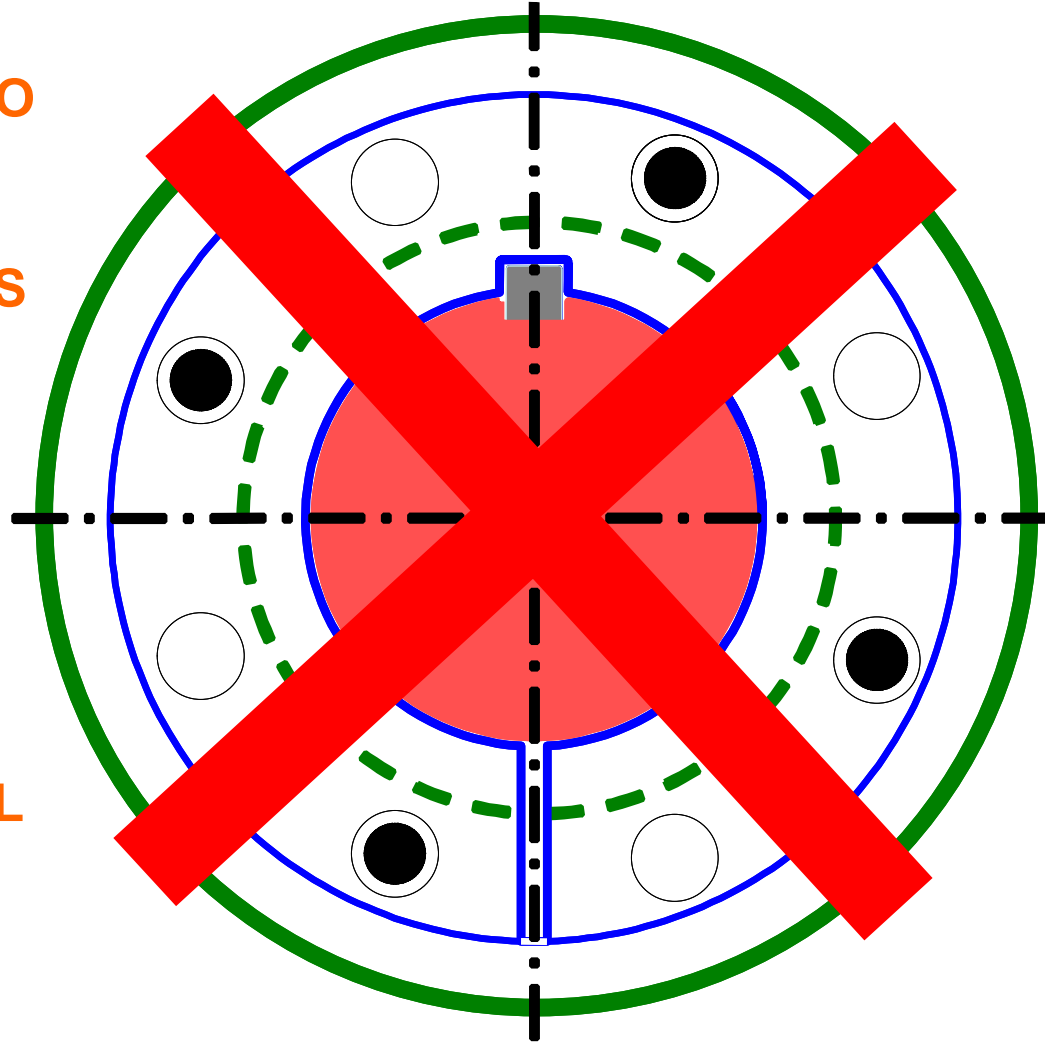
DISADVANTAGES OF...

REQUIRES HUB TO
HAVE MATCHING
TAPERS AND
MOUNTING HOLES

HIGH MACHINING
COST DUE TO
TAPER AND
DRILLING

ADDS BACKLASH
IN BIDIRECTIONAL
SHAFT/HUB
ROTATION

SINGLE POSITION
CAPABILITY



QD BUSHING SHAFT / HUB CONNECTIONS

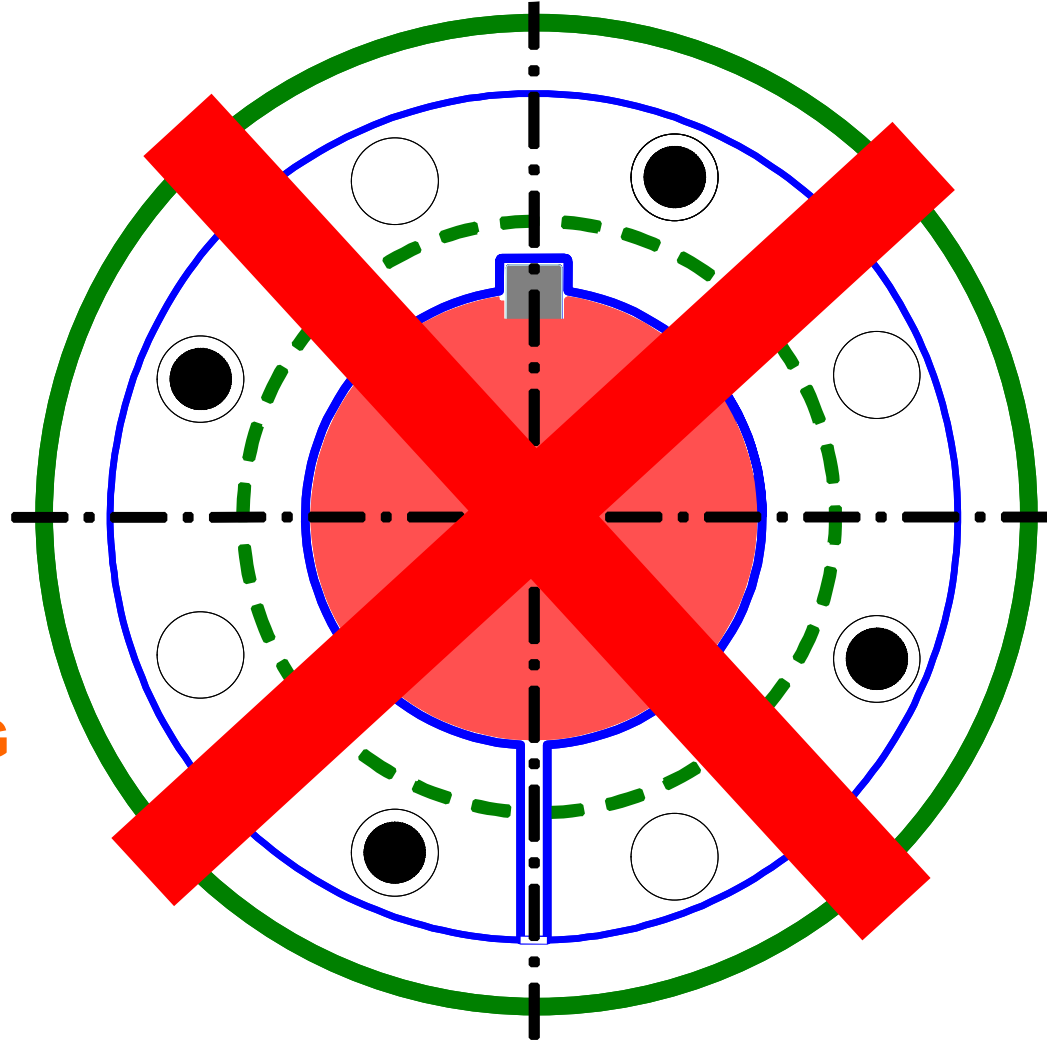
DISADVANTAGES OF...

REQUIRES
SECONDARY
LOCKING (e.g.
ASSEMBLY
SCREWS, KEY)

REQUIRES NON-
STANDARD
COMPONENTS

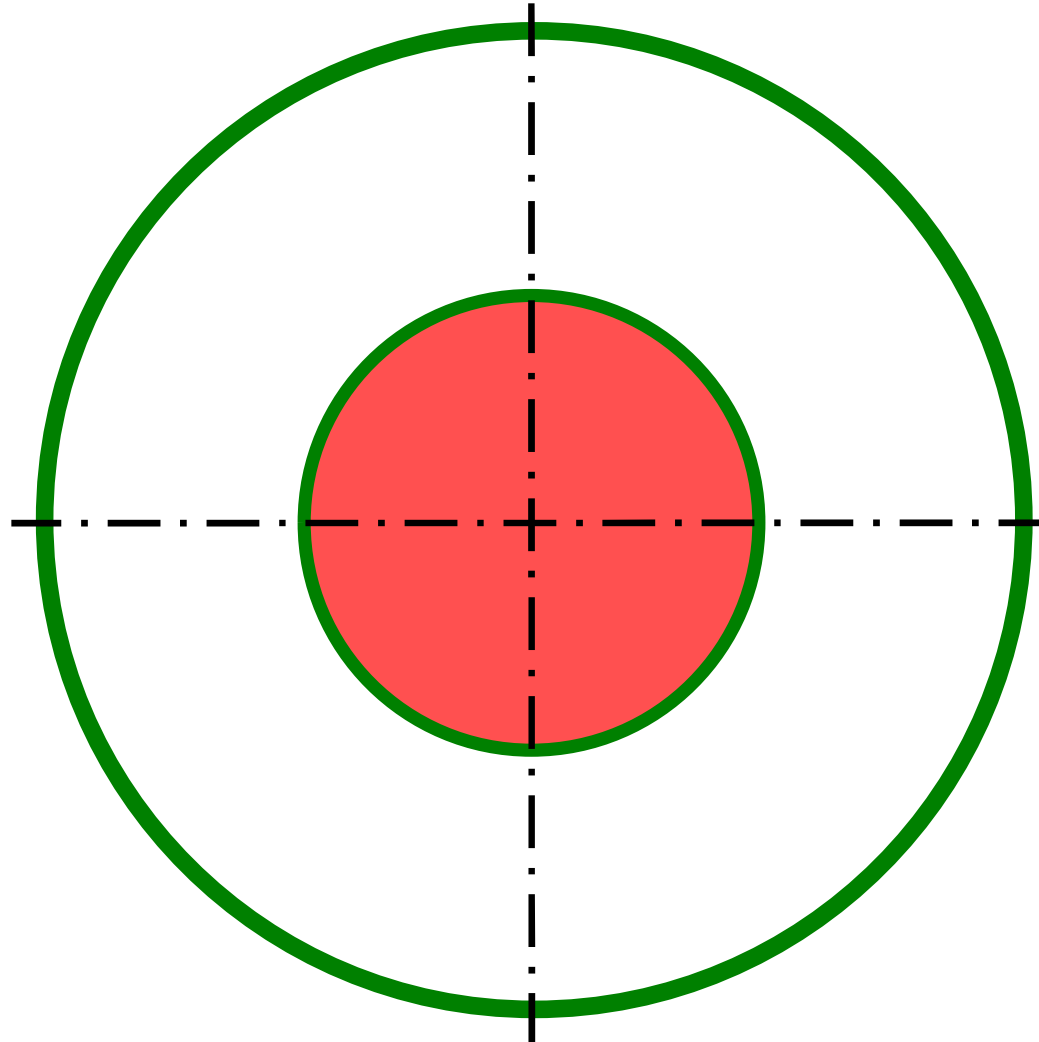
TIME CONSUMING
ASSEMBLY AND
DISASSEMBLY
PROCESS

LOCALIZED
STRESS
CONCENTRATION



QD BUSHING SHAFT / HUB CONNECTIONS

WHY NOT TO USE



INTERFERENCE FIT SHAFT / HUB CONNECTIONS

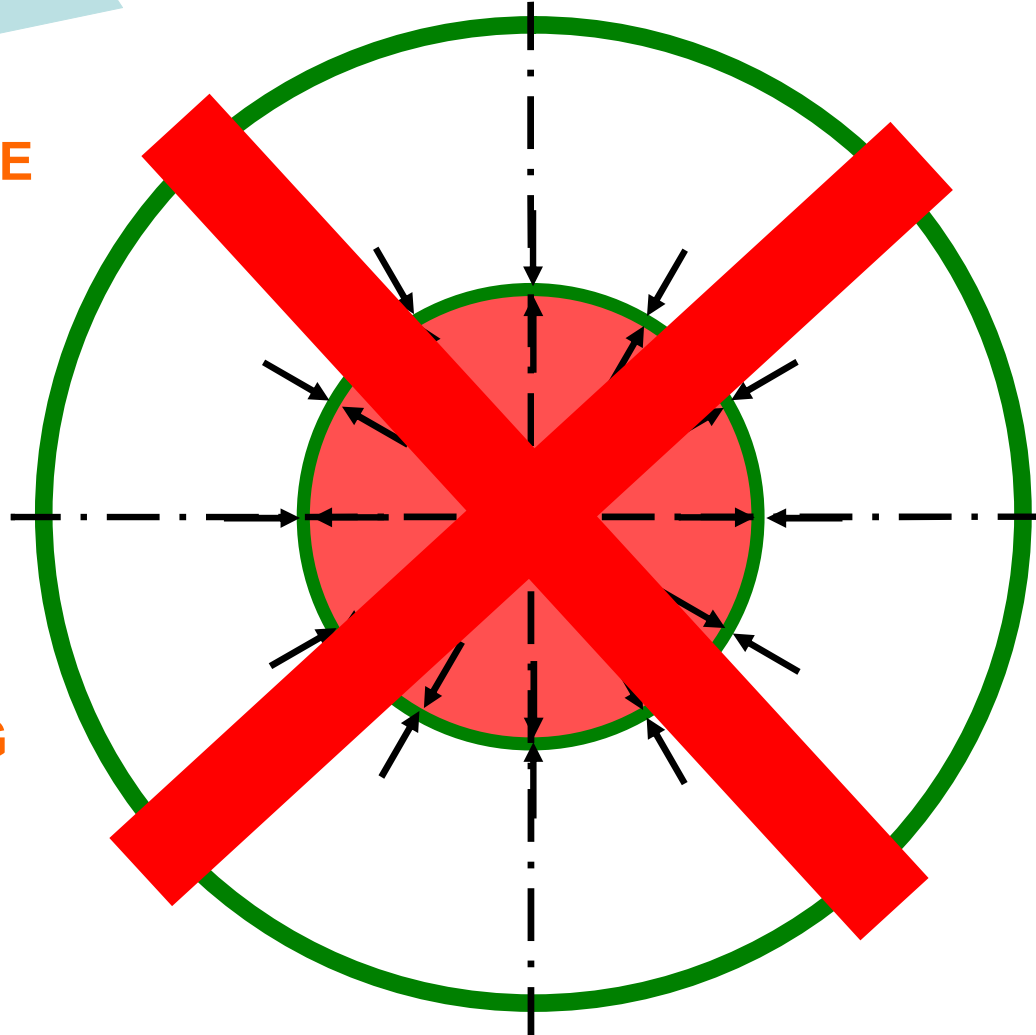
DISADVANTAGES OF...

REQUIRES
SPECIAL SURFACE
PREPARATION

EXTREMELY
DIFFICULT TO
DISASSEMBLE

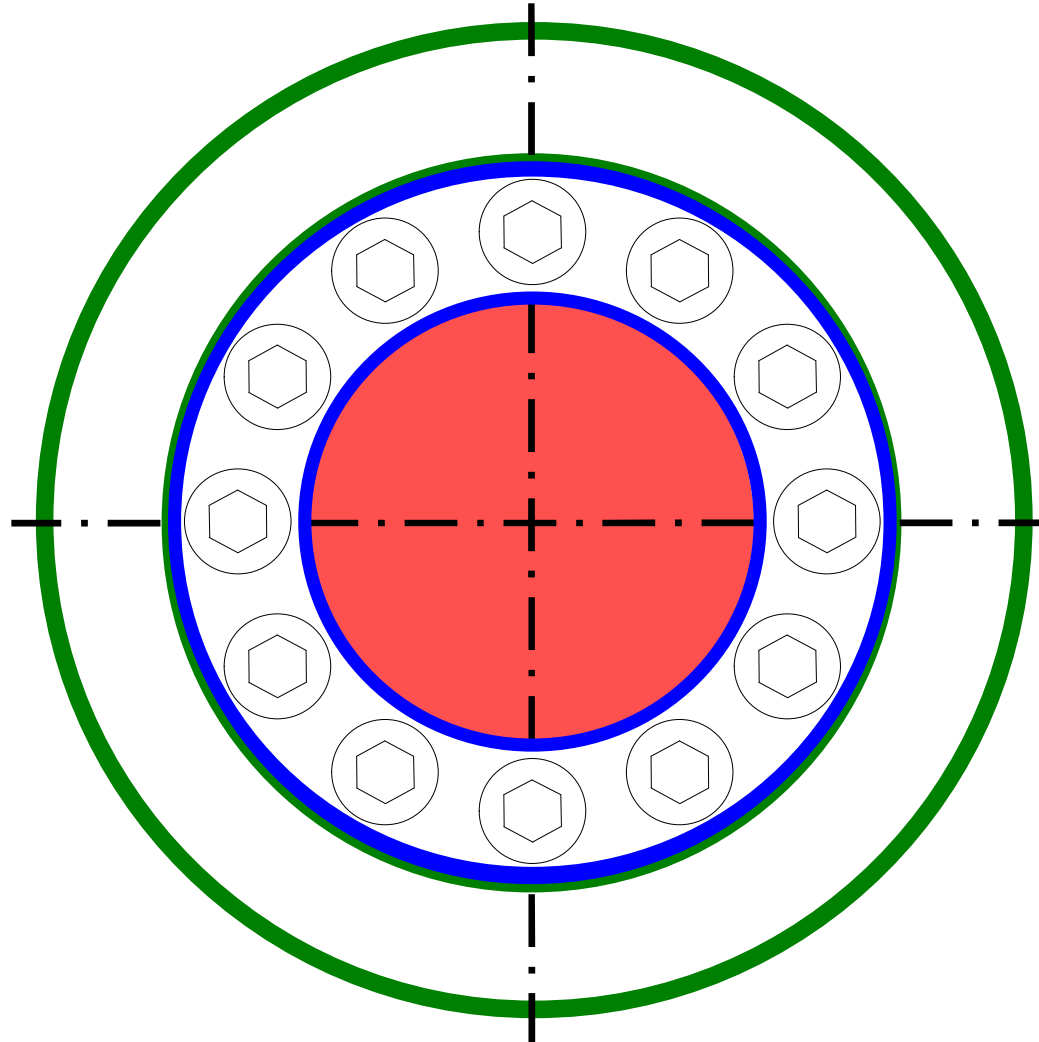
TIME CONSUMING
ASSEMBLY
PROCESS

SPECIAL
ASSEMBLY
TOOLING ARE
REQUIRED



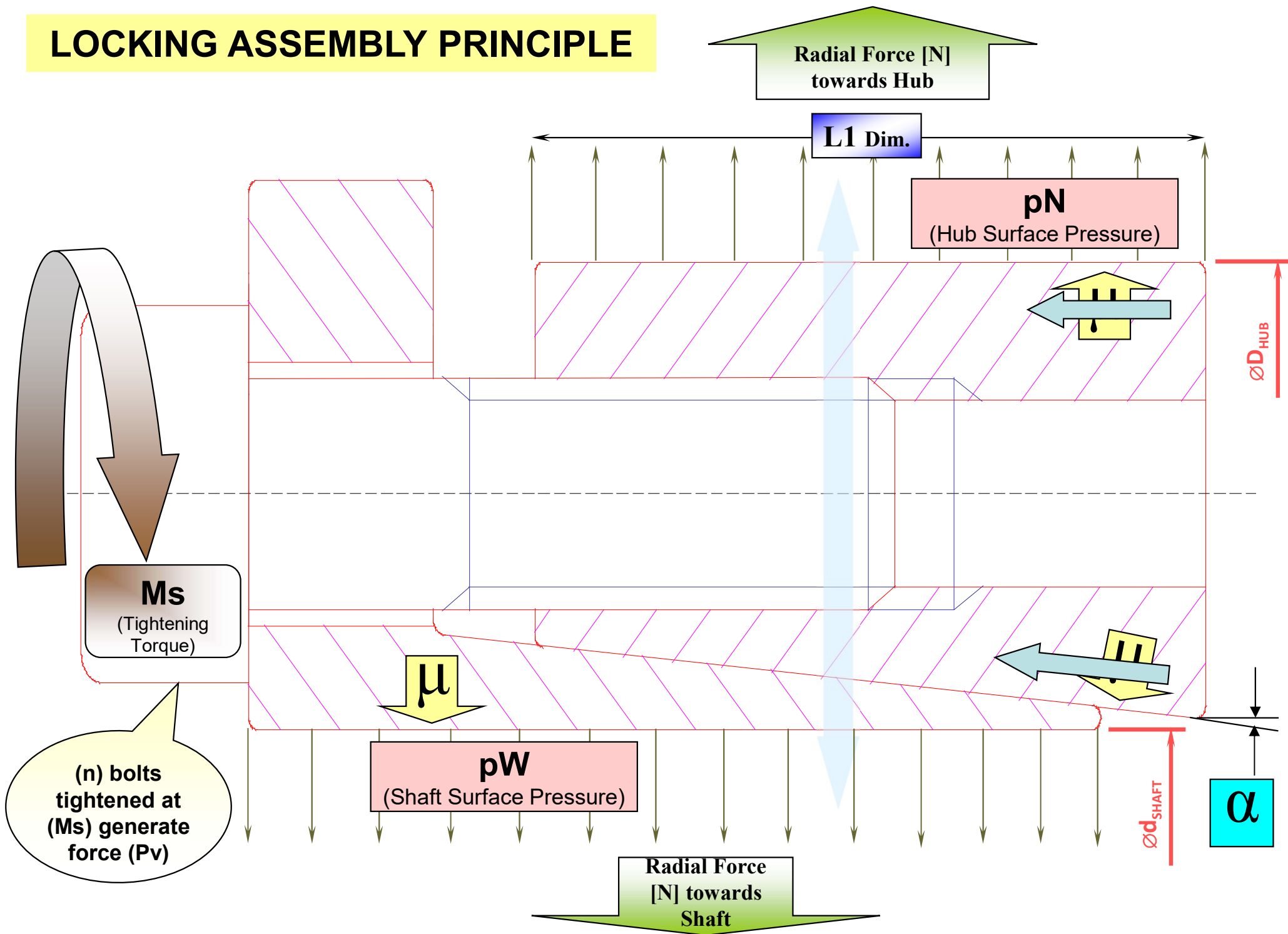
INTERFERENCE FIT SHAFT / HUB CONNECTIONS

ADVANTAGES OF INTERNAL TYPE



TOLLOK® SHAFT / HUB LOCKING ASSEMBLIES

LOCKING ASSEMBLY PRINCIPLE



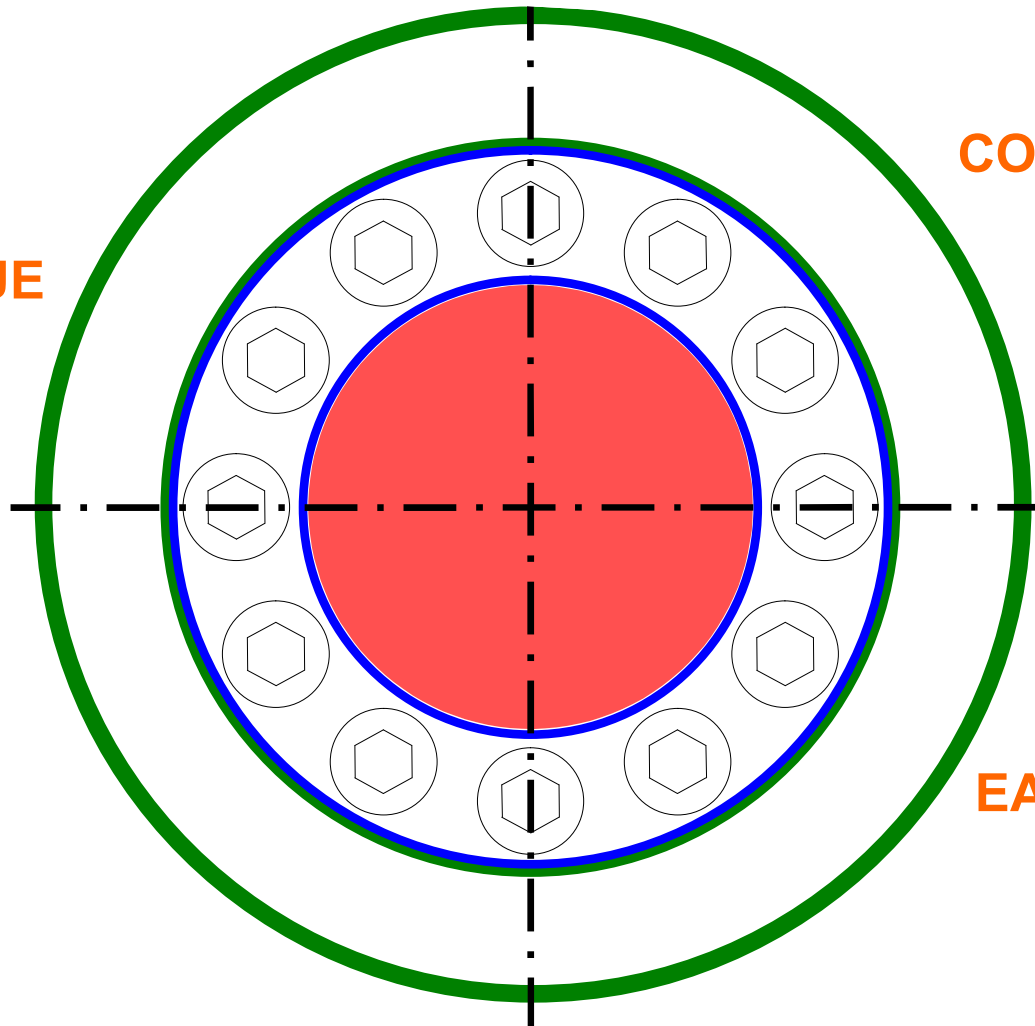
ADVANTAGES OF..

ENTIRE SHAFT &
HUB DIAMETERS
ARE USED TO
TRANSMIT TORQUE

FULL SURFACE
CONTACT PREVENTS
FRETTING
CORROSION

STRESSES ARE
SYMMETRICALLY
DISTRIBUTED
AROUND THE
CIRCUMFERENCE

EASY TO ASSEMBLE
& DISASSEMBLE



TOLLOK® SHAFT / HUB LOCKING ASSEMBLIES

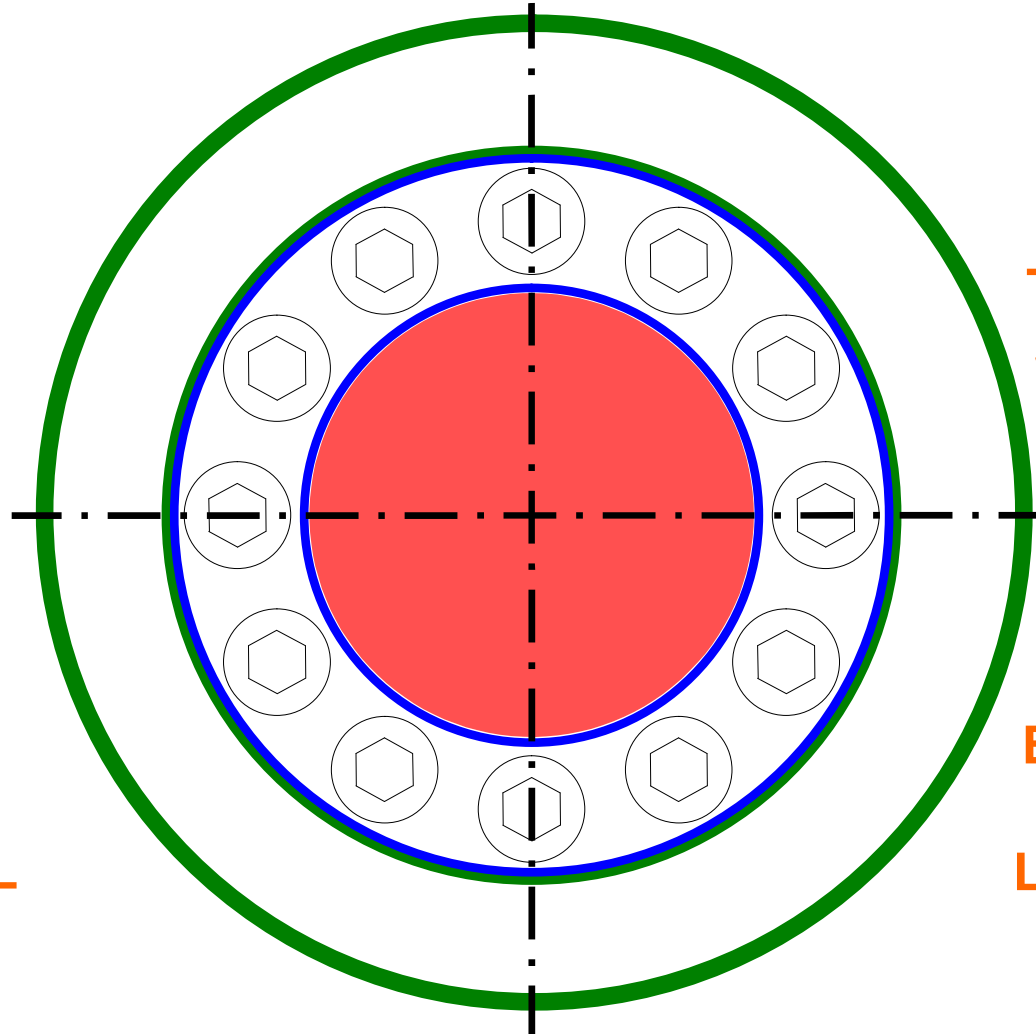
ADVANTAGES OF...

NO SECONDARY
OPERATIONS
REQUIRED (e.g.
KEYWAY, SPLINE,
ETC.)

REQUIRES
STANDARD
MACHINING
TOLERANCES FOR
SHAFT & HUB (e.g.
H8/h8)

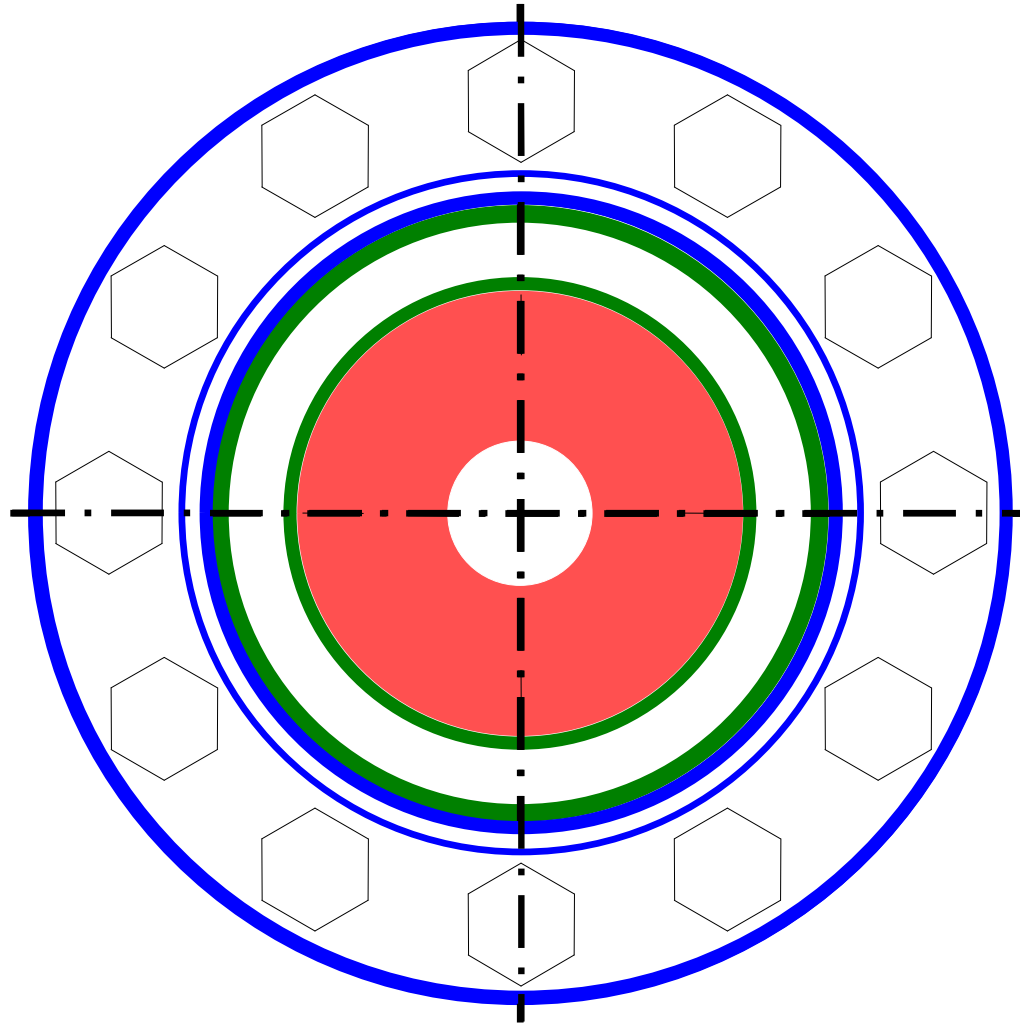
ZERO BACKLASH
IN BIDIRECTIONAL
ROTATION

EFFECTIVE UNDER
AXIAL THRUST
LOAD OR BENDING
MOMENTS



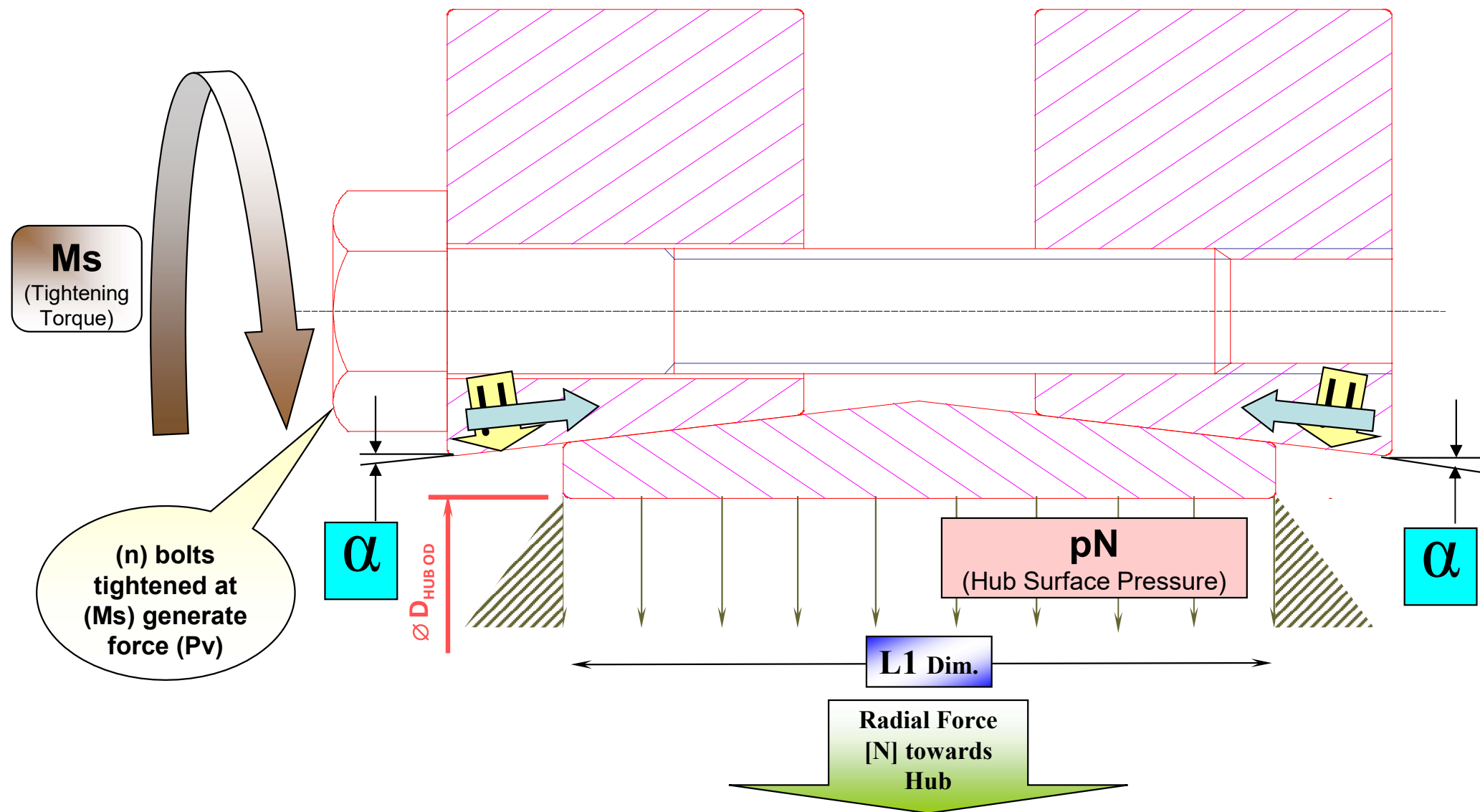
TOLLOK® SHAFT / HUB LOCKING ASSEMBLIES

ADVANTAGES OF SHRINK-DISC TYPE



TOLLOK® SHAFT / HUB LOCKING ASSEMBLIES

SHRINK-DISC PRINCIPLE



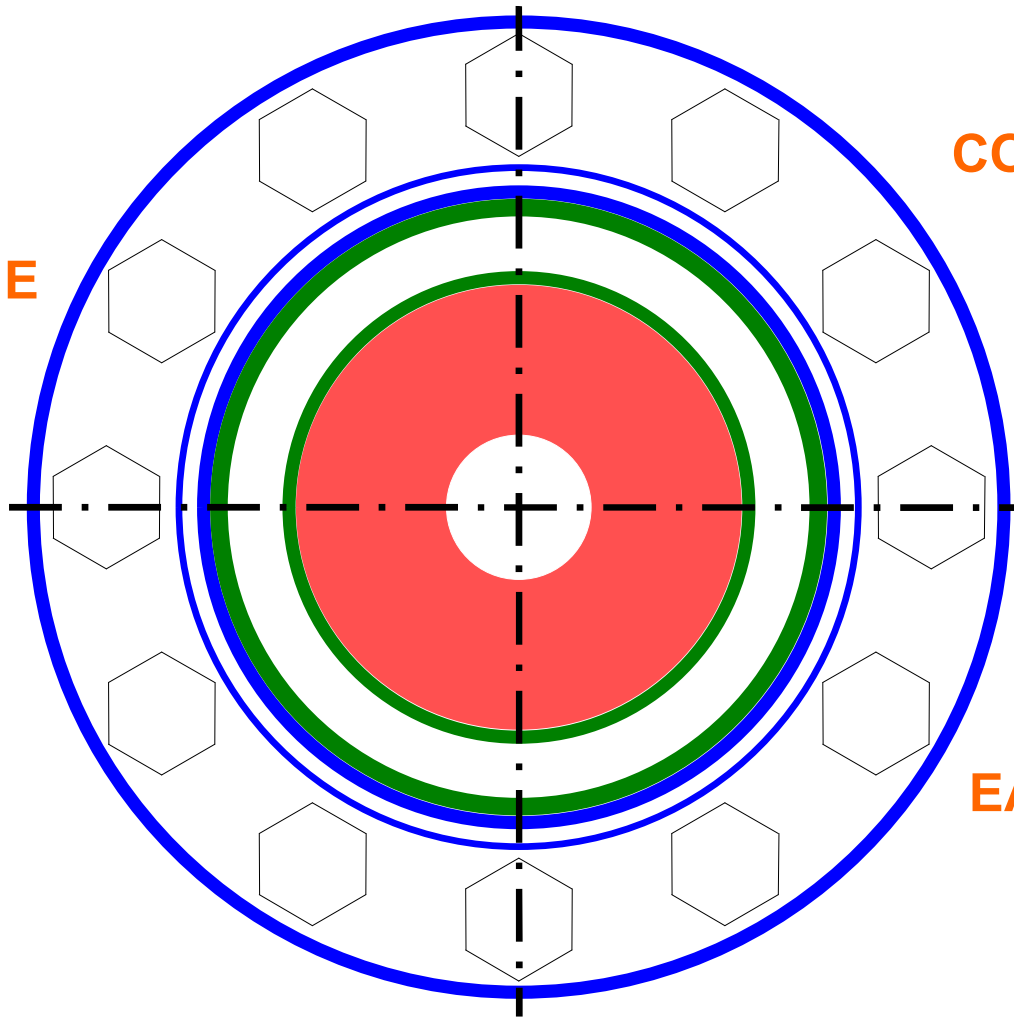
ADVANTAGES OF..

ENTIRE SHAFT &
HUB DIAMETERS
ARE USED TO
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FULL SURFACE
CONTACT PREVENTS
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STRESSES ARE
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EASY TO ASSEMBLE
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TOLLOK® SHAFT / HUB LOCKING ASSEMBLIES

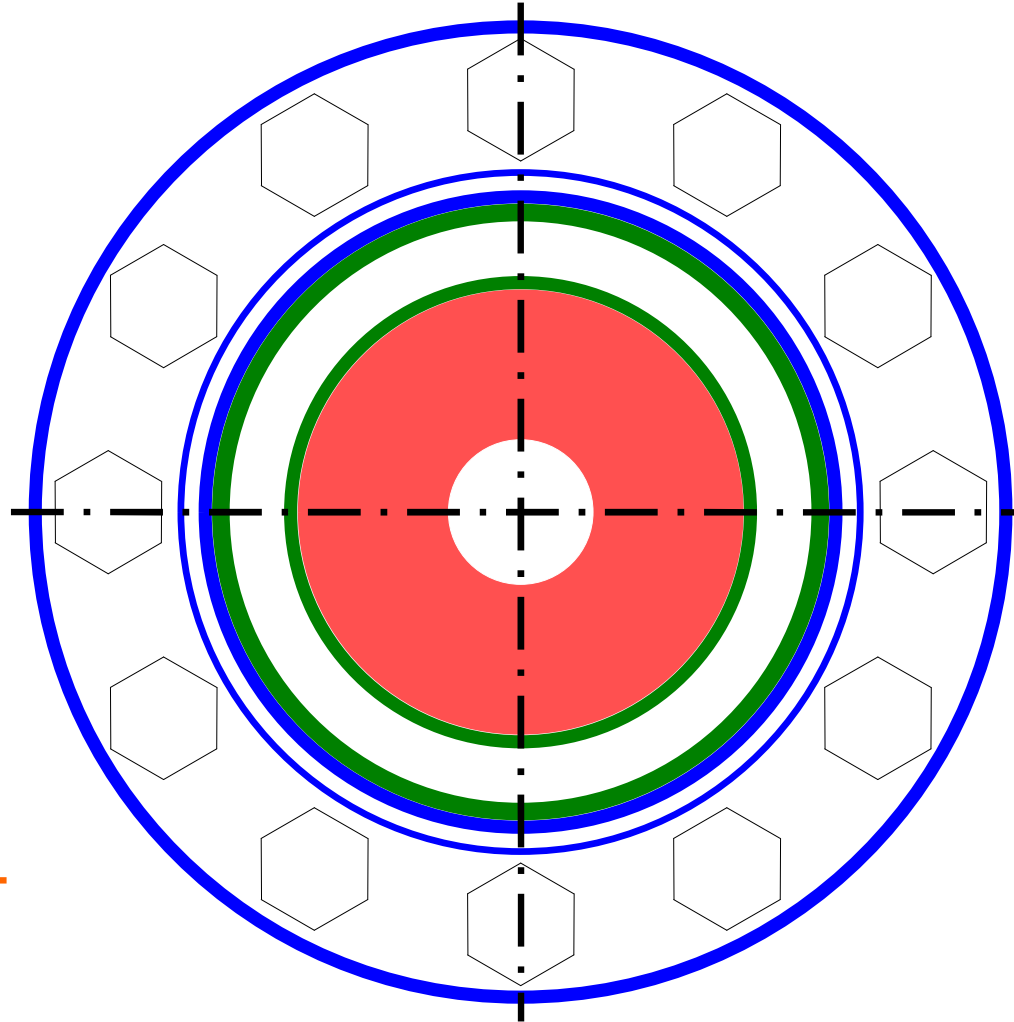
ADVANTAGES OF...

**NO SECONDARY
OPERATIONS
REQUIRED (e.g.
KEYWAY, SPLINE,
ETC.)**

**EXCELLENT
SOLUTION FOR
LONG HUBS (e.g.
HOLLOW SHAFT
GEAR REDUCERS)**

**ZERO BACKLASH
IN BIDIRECTIONAL
ROTATION**

**ONE UNIT CAN BE
USED WITH
MULTIPLE SHAFT
DIAMETERS**



TOLLOK® SHAFT / HUB LOCKING ASSEMBLIES

TOLLOK® LOCKING ASSEMBLIES



MODEL NO.: **TLK 132**



TLK 133



TLK 200



TLK 400



MODEL NO.: **TLK 250**



TLK 250L



TLK 500



TLK 60_

TOLLOK® APPLICATIONS



**Bevel Gear Set
Assembly**



**Machine
Spindle**



V-Belt Pulley



Drum Pulley



Timing Pulley



Index Wheel



**Pulley
Assembly**



TOLLOK®

KEYLESS

LOCKING ASSEMBLIES